

Question:

What is the de Broglie wavelength of an electron with a kinetic energy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think> De Broglie wavelength is given by the formula $\lambda = \frac{h}{p}$, where (h) is Planck's constant and (p) is momentum. Momentum is given by $p = \sqrt{2mE}$, where (m) is mass, and (E) is energy. For an electron, mass $(m = 9.11 \times 10^{-31} \text{ kg})$. The energy in electron volts is $1 \text{ eV} = (1.6 \times 10^{-19} \text{ J})$. First, convert energy to joules. Next, calculate momentum. Then, use this to find wavelength. </think>

<answer> A. 2.48×10^{-8} cm </answer>

